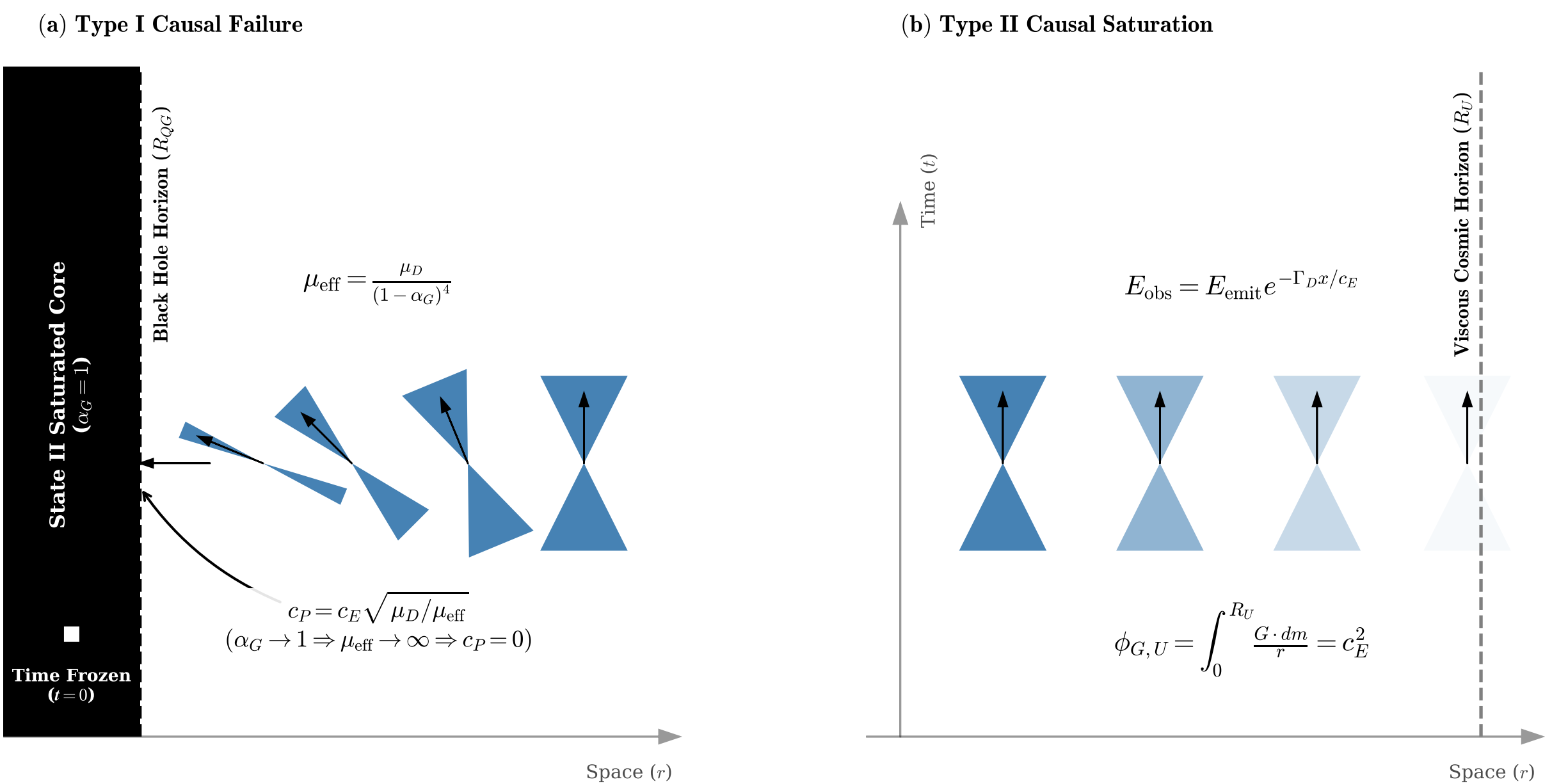

(a) Type I Causal Failure
State II Saturated Core
($\alpha_G = 1$)
Black Hole Horizon (R_{QG})
$\mu_{\text{eff}} = \frac{\mu_D}{(1-\alpha_G)^4}$
$c_P = c_E\sqrt{\mu_D/\mu_{\text{eff}}}$
$(\alpha_G \to 1 \Rightarrow \mu_{\text{eff}} \to \infty \Rightarrow c_P = 0)$
Time Frozen
($t = 0$)
Space (r)
(b) Type II Causal Saturation
Time (t)
Viscous Cosmic Horizon (R_U)
$E_{\text{obs}} = E_{\text{emit}} e^{-\Gamma_D x/c_E}$
$\phi_{G,U} = \int_0^{R_U} \frac{G \cdot dm}{r} = c_E^2$
Space (r)